

A Simple Approach to Crystal Field Theory

An introduction to the concepts of crystal field theory for beginning chemistry students has been presented in a number of recent sources.¹ This article presents a different approach to crystal field theory which should be particularly helpful to beginning students.

Crystal field theory relates the shape of an atom or ion to the arrangement of the atoms or ions which surround it in a crystal or molecule. The basic concepts of crystal field theory can be visualized most easily by considering atoms or ions which contain partially filled p subshells. In practice, however, the theory has the vast majority of its applications in atoms which have partially filled d subshells. The first two examples chosen therefore illustrate principles but involve molecules which are unlikely.

A simple crystal field model includes the following simplifications: The shape of an atom in a molecule is identical to its shape as an isolated atom.² Atoms are hard and have volume; two cannot occupy the same space. The atoms or ions surrounding the central atom (ligands) will approach the atom in such a fashion as to come as close as possible to the nucleus of the central atom. This occurs because the ligands are usually negative ions or polar molecules and are attracted to the center of positive charge.

To illustrate this crystal field model consider the formation of such an unlikely species as a complex between a silicon atom and fluoride ions. An isolated silicon atom has the shape of a flattened sphere (see Fig. 1). The fluoride ions would reside on the flattened sides of the sphere, since in these positions they are nearer the nucleus. Since only one F^- ion fits on each flattened side of the silicon atom, this simple approach predicts a formula SiF_2^{-2} for the complex and a linear structure. Other fluoride ions could be more loosely bound around the equator of the molecule. A further consequence of the theory involves the energies of the three $3p$ orbitals of Si. The flattened spheroidal shape of silicon arises from the presence of p electrons in only two of the three p orbitals. In SiF_2^{-2} the fluoride ions lie on the axis of the empty p orbital. In this complex ion silicon electrons would be repelled by the fluoride ions on the one axis and would remain in orbitals oriented along other axes. An alternative way of making this statement is to say the p orbital on the axis occupied by the two fluoride ions will have a higher

energy than the other two p orbitals. In an isolated Si atom all three $3p$ orbitals are equivalent in energy and any two may be occupied. The formation of the SiF_2^{-2} complex increases the energy of the $3p$ orbital facing the F^- ions and causes the two $3p$ electrons to remain in the other two $3p$ orbitals (see Fig. 2).

As a second illustration of crystal field theory let us predict the geometry and coordination number of another unlikely molecule ClF_n^{+4-n} . An isolated Cl^{+4} ion has the shape of an elongated sphere (Fig. 1).

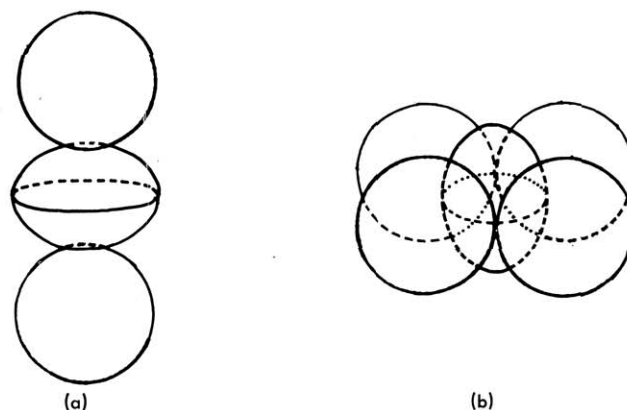


Figure 1. (a) Two fluoride ions arranged around a silicon atom. (b) Four fluoride ions around a Cl^{+4} ion.

The fluoride ions would be attracted around the equator of this Cl^{+4} ion in the plane of the two empty p orbitals, since in this plane they can approach more closely the positive Cl^{+4} nucleus. The coordination number (n) of the Cl^{+4} ion will depend on the number of fluoride ions which can fit around the equator of the spheroidal Cl^{+4} ion. A reasonable value is four (a complex of Cl^{+3} , ClF_4^- , does exist and has a planar structure). In ClF_4 one p orbital is lower in energy than the other two. This results because the four fluoride ions lie in the plane of two p orbitals and repel electrons from them (Fig. 3).

These two examples illustrate two basic features of crystal field theory: that the coordination number of an

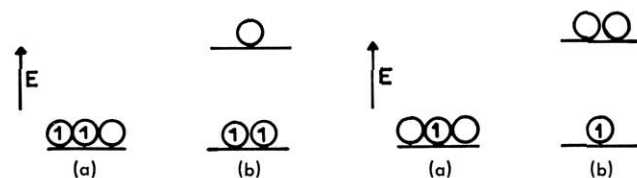


Figure 2 (left). The relative energies of p orbitals (a) in an isolated Si atom and (b) in a silicon atom in a hypothetical linear complex SiF_2^{-2} .

Figure 3 (right). The relative energies of p orbitals (a) in an isolated Cl^{+4} ion and (b) in a Cl^{+4} ion in the compound ClF_4 .

¹ COTTON, F. A., J. CHEM EDUC., **41**, 466 (1964); BASOLO, F., AND JOHNSON, R. C., "Coordination Chemistry," W. A. Benjamin, New York, 1964, pp. 34-49.

² Shapes of atoms are discussed in JOHNSON, R. C., and RETTEW, R. R., THIS JOURNAL **42**, 145 (1965).

atom or ion and the arrangement of other atoms around the central atom depend on the number and kind of electrons present in that central atom (i.e., the shape of the central atom); and that the presence of atoms around a central atom causes orbitals which were of equal energy in the isolated central atom to have different energies.

Let us now consider one example of crystal field concepts as applied to ions containing partially filled d subshells. The Cr^{+3} ion has the electronic configuration $[\text{Ar}]3d^3$. If one assumes the three electrons are in the $3d_{xy}$, $3d_{xz}$, and $3d_{yz}$ orbitals, the shape of this ion arises essentially from the electron geometry of these three d orbitals. The probability of finding a $3d_{xy}$, $3d_{xz}$, or $3d_{yz}$ electron is zero on the x , y , and z axes.

Therefore, anions seeking a positive Cr^{+3} ion would approach along these axes. One might predict therefore that the fluoride complex of Cr^{+3} would contain six fluoride ions (CrF_6^{-3}) with these ions positioned one on each side of the chromium ion on the x , y , and z axes. In fact all Cr^{+3} complexes have this octahedral geometry. One might further guess that CrX_6^{-3} would be slow to react, since an ion or molecule seeking to approach the positive Cr^{+3} ion would be repelled by the X^- groups on the x , y , and z axes and by the d_{xy} , d_{xz} , and d_{yz} electrons which spend their time between the axes. On the other hand the X^- groups bonded to the Cr^{+3} will be held firmly since they are excellently oriented to "see" the positive charge of the Cr^{+3} nucleus. Slow reactions are characteristic of Cr^{+3} complexes.

